

CONFIDENTIAL



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

SCHOOL OF COMPUTING
Faculty of Engineering

UNIVERSITI TEKNOLOGI MALAYSIA

FINAL EXAMINATION

SEMESTER II 2020/2021

SUBJECT CODE : SCS12143/SECI2143/SECI1143

SUBJECT NAME : PROBABILITY & STATISTICAL DATA ANALYSIS

YEAR/COURSE :

TIME : 3 HOURS

DATE :

INSTRUCTIONS TO THE STUDENTS:

1. Please answer ALL the questions in the answer sheet form.
2. Fill in your particular in the answer sheet.
3. Do calculations in 3 decimal places.

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SECTION	
LECTURER	

(This question paper consist of 8 pages, including this pages)

QUESTION 1**[10 MARKS]**

- a) In a study of 31950000 (31.95 million) for a country's population, 533000 Covid-19 cases have been detected within 15 months. Construct a 95% confidence interval for the proportion of people that confirms Covid-19.

(4 marks)

- b) In a study of all residents from City A, sixty Covid-19 cases have been detected. Among these confirmed cases, most of them have the symptoms of fever. The following temperatures (°C) of a resident were collected within fourteen days of self-quarantine.

36.3	37.5	35.8	36.6	36.2	38.6	38.4
37.9	35.8	38.0	38.3	37.1	37.2	36.5

With the use of t -test to perform the hypothesis testing at the 95% significance level, do the data provide sufficient evidence to conclude that the average temperature of all residents from City A is more than 37.0 °C?

(6 marks)

QUESTION 2**[20 MARKS]**

- a) Company A and Company B are two digital film producers. A study was conducted to determine if there is a significant difference in terms of running time (minutes) of montage video produce by them. The collected data as shown below.

Company A	Company B
102	81
86	165
98	97
109	134
92	92
87	
114	

Test the hypothesis at level of significance, $\alpha = 0.1$ that $\sigma_1^2 = \sigma_2^2$ against the alternative that $\sigma_1^2 > \sigma_2^2$, where σ_1^2 and σ_2^2 are the variances for the running times of montage video produced by Company A and Company B, respectively. Assume the distribution of times to be approximately normal.

(7 marks)

- b) The following data give the times of 15 girls when they were put on ultraheavy rope-jumping program. Someone thought that such a program would increase their speed in the 40-yard dash. Let d equal the different in time run the 40-yard dash – the “before program time (x)” minus the “after-program time (y)”. Assume that the distribution of d is approximately normal, $N(\mu_d, \sigma_d^2)$. Test the null hypothesis $H_0: \mu_d = 0$ against the alternative hypothesis $H_1: \mu_d > 0$. Use $\alpha = 0.05$

x	y
24.97	23.98
22.93	22.83
22.38	22.39
22.76	21.66
22.98	23.17
24.76	23.63
23.41	22.75
22.91	22.34
21.82	21.68
23.80	22.53
23.80	23.61
22.10	22.74
22.08	21.95
21.80	21.58
22.84	23.58

(13 marks)

QUESTION 3**[20 MARKS]**

- a) A TV host claims that backpackers' favourite locations vary according to the age of the travellers. He claimed that the distribution for the travel age is as follows: 20-29 years old (16%), 30-39 years old (40%), 40-49 years old (25%) and 50 and over (19%). To prove this, his friends interviewed 280 backpackers. The backpackers were asked as to where they would normally travel. Table 1 shows the result of the interview based on age group.

Table 1

20-29	30-39	40-49	50 and over
60	100	80	40

Conduct the hypothesis testing with 95% confidence level to test the claim by the TV hosts. Use notation as follows: p_1 = 20-29 years old travellers, p_2 = 30-39 years old travellers, p_3 = 40-49 years old travellers and p_4 = 50 years old traveller.

(10 marks)

- b) A researcher wishes to test the effectiveness of new flu vaccination. There are 150 people who are vaccinated with Vaccine A, 180 are vaccinated with Vaccine B and 100 people are not vaccinated. Independent simple random samples were used and the number in each group who later caught the flu was recorded. Table 2 show the test group and the flu status.

Table 2

	Vaccine A	Vaccine B	No Vaccine
Caught the flu	8	19	21
Did not catch the flu	142	161	79

At the 0.05 level of significance, is the new flu vaccination effective? Conduct the chi-square test to support your justification.

(10 marks)

QUESTION 4**[15 MARKS]**

A biscuit production manager wants to investigate the linear relationship between the number of items produced and the production cost. To pursue his objective, the manager recorded the data on the number of items produced and the production cost per day (in thousands of Malaysian Ringgit) as shown at Table 3.

Table 3

Days	1	2	3	4	5	6	7	8	9	10	11	12
Number of items, (x)	26	44	53	29	77	80	20	40	67	86	17	61
Production cost, (y)	42	60	69	47	91	98	39	55	85	104	37	77

- i) Name the **TWO (2)** most popular correlation coefficient.
(2 marks)
- ii) Based on the data in Table 3, calculate the correlation coefficient, r .
(11 marks)
- iii) Based on correlation coefficient, r obtained, make a conclusion on the linear relationship between the number of items produced and the production cost.
(2 marks)

- i) Pearson Product-Moment Correlation
- Spearman's Rho Rank correlation

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$$r = \frac{\sum xy - \frac{(\sum x \sum y)}{n}}{\sqrt{\left(\sum x^2 - \frac{(\sum x)^2}{n}\right)\left(\sum y^2 - \frac{(\sum y)^2}{n}\right)}}$$

$$1) r = \frac{\sum xy - \frac{(\sum x \sum y)}{n}}{\sqrt{(\sum x^2 - \frac{(\sum x)^2}{n})(\sum y^2 - \frac{(\sum y)^2}{n})}}$$

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Production cost, (y)	42	60	69	47	91	98	39	55	85	104	37	77
Σx	1092	2640	3667	1368	7007	7840	780	2200	5605	8944	629	4697
Σx^2	676	1936	2809	841	5929	6400	400	1600	4489	7396	289	3721
Σy	42	60	69	47	91	98	39	55	85	104	37	77
Σy^2	1764	3600	4761	1764	8100	9604	1521	3025	7225	10816	1369	5929
Σxy	1092	2640	3667	1368	7007	7840	780	2200	5605	8944	629	4697
Σx^3	27	8	27	64	125	216	343	512	729	1000	1331	1728
Σy^3	72	216	343	91	673	784	152	302	595	1064	523	377

600 * Use calculator for the
804 fastest answer!!

value from calc:

$$\begin{aligned}\Sigma x &= 600 \\ \Sigma x^2 &= 36486 \\ \Sigma y &= 804 \\ \Sigma y^2 &= 60104 \\ \Sigma xy &= 46544\end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad r &= \frac{46544 - \frac{(600)(804)}{12}}{\sqrt{\left((36486) - \frac{(600)^2}{12} \right) \left((60104) - \frac{(804)^2}{12} \right)}} \\ &= 0.9975 \end{aligned}$$

iii) because the correlation coefficient, $r = 0.995$ is really close to 1, there is a strong positive linear relationship between number of items and production cost. As number of items increases, the production cost increase along with it.

QUESTION 5**[15 MARKS]**

Last year, five randomly selected students took a math aptitude test before they began their statistic course. The Statistics Department would like to analyze the relationship of the data, make prediction using the regression equation and validate the regression equation. The data of the math aptitude test score and corresponding statistics grade are as shown in Table 4 below.

Table 4

Score on math aptitude test (x)	Statistics grade (y)
95	85
85	95
80	70
70	75
65	70

- i) Calculate $\sum x$, $\sum y$, $\sum xy$ and $\sum x^2$ (4 marks)
- ii) Calculate the value of b_1 and b_0 (2 marks)
- iii) Based on answers in (i) and (ii), what linear regression equation best predicts statistics performance, based on math aptitude scores? (3 marks)
- iv) If a student made a 60 on the aptitude test, what grade would we expect her to make in statistics? (2 marks)
- v) Find value of SSR , SST and R^2 . (3 marks)
- vi) Based on your answer in (v), how well does the regression equation fit the data? (1 mark)

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(2 marks)
- v) Find value of SSR , SST and R^2 .

(3 marks)
- vi) Based on your answer in (v), how well does the regression equation fit the data?

(1 mark)

ii) $\hat{y} = 31.18 + 0.6053(60)$
 $= 67.50$

v) $SSR = SSE + SST$

$SST = \sum (y - \bar{y})^2$
 $SSE = \sum (y - \hat{y})^2$

$SST = 470$

$SSR = ((31.18 + 0.6053(95)) - 79)^2 + \dots + \dots$
 $= 208.83$

$R^2 = \frac{SSR}{SST}$
 $= \frac{208.83}{470}$
 $= 0.4443$

iv) $R^2 = 0.4443$ approximately 44% of the variation in statistics grades explain by math aptitude test. This indicates that moderate to weak linear fit. as more than 55% of the variation remain unexplain.

i) $\sum x = 95 + 85 + 80 + 70 + 65$
 $= 395$

$\sum y = 85 + 95 + 70 + 75 + 70$
 $= 395$

$\sum xy =$
 $= 31550$

$\sum x^2 = 95^2 + 85^2 + 80^2 + 70^2 + 65^2$
 $= 31775$

ii) $b_1 =$

$$\hat{y} = b_0 + b_1x$$
$$b_1 = \frac{\sum xy - \frac{\sum x \sum y}{n}}{\sum x^2 - \frac{(\sum x)^2}{n}}$$
$$b_0 = \bar{y} - b_1\bar{x}$$

$b_1 = \frac{31550 - \frac{395(395)}{5}}{31775 - \frac{(395)^2}{5}}$
 $= 0.6053$

$b_0 = \bar{y} - b_1\bar{x}$
 $= 79 - 0.6053(79)$
 $= 31.18$

iii) linear regression equation

$\hat{y} = b_0 + b_1x$
 $\hat{y} = 31.18 + 0.6053x$

QUESTION 6**[20 MARKS]**

Table 5 shows the data regarding the overall results for statistics class for FIVE (5) randomly selected students from each of the THREE (3) batches. Does this data provide evidence that there is any significant difference in the results throughout all the batches? Test the appropriate hypothesis using 95% significance level.

Table 5

Batch 1	Batch 2	Batch 3
78.0	81.5	83.5
80.0	77.3	82.1
67.5	59.3	78.4
74.3	78.1	65.4
88.0	83.4	77.1

(20 marks)